

Final Report: Association Rule Mining on Global Pollution and Energy Data

Apriori Algorithm, FP-Growth Comparison, Chi-Square Validation, Sensitivity Analysis, and Country-Level Recommendations

Metric	Final Value
Dataset size	200 rows (from supplied Global Pollution Analysis dataset)
Minimum support	0.05
Minimum confidence	0.60
Frequent itemsets	566
Association rules	34
Strong significant rules	34
Average confidence	0.6453
Average lift	1.9355
Maximum lift	2.7137

1. Introduction

The objective of this project is to identify meaningful relationships among pollution severity, industrial waste, CO2 emissions, energy consumption, energy recovery, and renewable energy usage. The assigned task is an association rule mining problem, so the Apriori algorithm was applied instead of regression or classification. FP-Growth was also used as a comparison method to check whether the discovered rules are consistent across algorithms.

2. Dataset and Preprocessing

The dataset contains pollution and energy-related indicators for countries. Numerical columns were cleaned and converted into categorical levels: Low, Medium, and High. This binning step was required because association rule mining works with transactional or categorical itemsets rather than raw continuous numerical values. The resulting categorical dataset was one-hot encoded into Boolean columns for Apriori and FP-Growth.

```
# Correct binning approach used in preprocessing

def percentile_bin(series):
    return pd.qcut(series, q=3, labels=["Low", "Medium", "High"], duplicates="drop")

category_cols = [
    'Air_Pollution_Severity', 'Water_Pollution_Severity',
    'Soil_Pollution_Severity', 'Industrial_Waste_Level',
    'CO2_Emission_Level', 'Energy_Consumption_Level',
    'Energy_Recovery_Level', 'Renewable_Energy_Level'
]

transaction_df = df[category_cols].copy()
encoded_df = pd.get_dummies(transaction_df).astype(bool)
```

3. Methodology

Apriori was used to generate frequent itemsets with minimum support of 0.05. Association rules were then generated with minimum confidence of 0.60 and filtered to retain only rules with lift greater than 1.0. A corrected Chi-Square test was applied to validate full multi-item antecedents and consequents. Rules with p-values less than 0.05 were treated as statistically significant.

```
MIN_SUPPORT = 0.05
MIN_CONFIDENCE = 0.60

frequent_itemsets = apriori(encoded_df, min_support=MIN_SUPPORT, use_colnames=True)
rules = association_rules(frequent_itemsets, metric='confidence', min_threshold=MIN_CONFIDENCE)
```

```

rules = rules[rules['lift'] > 1.0]

# Multi-item Chi-Square validation
ants = list(rule['antecedents'])
cons = list(rule['consequents'])
ant_true = encoded_df[ants].all(axis=1)
con_true = encoded_df[cons].all(axis=1)
contingency = [[(ant_true & con_true).sum(), (ant_true & ~con_true).sum()),
               [ (~ant_true & con_true).sum(), (~ant_true & ~con_true).sum()]]
chi2, p, dof, expected = chi2_contingency(contingency)

```

4. Evaluation Metrics

Support measures how frequently a rule pattern appears in the dataset. Confidence measures the reliability of a rule. Lift measures how much stronger the relationship is compared to random chance. The p-value from the Chi-Square test measures whether the rule association is statistically significant.

5. Key Findings

The final model produced 34 strong statistically significant rules. All selected rules satisfy confidence ≥ 0.60 , lift > 1.0 , and p-value < 0.05 . The average confidence of the strong rules is 0.6453, and the average lift is 1.9355, indicating that the discovered patterns are reliable and stronger than random chance.

Top 5 Strong Rules by Lift

#	Antecedent	Consequent	Support	Conf.	Lift	p-value
1	Renewable_Energy_Level_Medium, Industrial_Waste_Level_Low, Air_Pollution_Severity_High	Soil_Pollution_Severity_Low	0.050	0.909	2.714	0.0001
2	Soil_Pollution_Severity_Low, Industrial_Waste_Level_Low, Air_Pollution_Severity_High	Renewable_Energy_Level_Medium	0.050	0.769	2.331	0.0015
3	Soil_Pollution_Severity_Low, Renewable_Energy_Level_Medium, Industrial_Waste_Level_Low	Air_Pollution_Severity_High	0.050	0.769	2.296	0.0018
4	Energy_Recovery_Level_High, Water_Pollution_Severity_Medium	CO2_Emission_Level_Low	0.055	0.688	2.052	0.0045
5	Renewable_Energy_Level_High, Air_Pollution_Severity_Low	Industrial_Waste_Level_Low	0.060	0.667	1.990	0.0042

Interpretation of the strongest rule: records with medium renewable energy usage, low industrial waste, and high air pollution are associated with low soil pollution. Its confidence of 90.91% means that the consequent appears in most cases where the antecedent appears. The lift of 2.7137 shows that the relationship is much stronger than random chance, and the p-value of 0.0001 confirms statistical significance.

6. Algorithm Comparison

FP-Growth was used as a second frequent-pattern mining method. This comparison is useful because Apriori and FP-Growth use different mining strategies. When both methods return the same or highly similar rule metrics, the patterns are less likely to be artifacts of one algorithmic approach.

```

fp_itemsets = fpgrowth(encoded_df, min_support=MIN_SUPPORT, use_colnames=True)
fp_rules = association_rules(fp_itemsets, metric='confidence', min_threshold=MIN_CONFIDENCE)
fp_rules = fp_rules[fp_rules['lift'] > 1.0]

print('Apriori rules:', len(rules))
print('FP-Growth rules:', len(fp_rules))

```

7. Sensitivity Analysis

The sensitivity analysis shows that lower support thresholds generate many rules, while higher support thresholds generate fewer but more reliable rules. At support 0.03 and confidence 0.60, the model produced 515 rules. At support 0.05 and confidence 0.60, it produced 34 rules. At support 0.10 and

confidence 0.60, it produced no rules. Therefore, support 0.05 and confidence 0.60 provide a balanced and interpretable threshold setting.

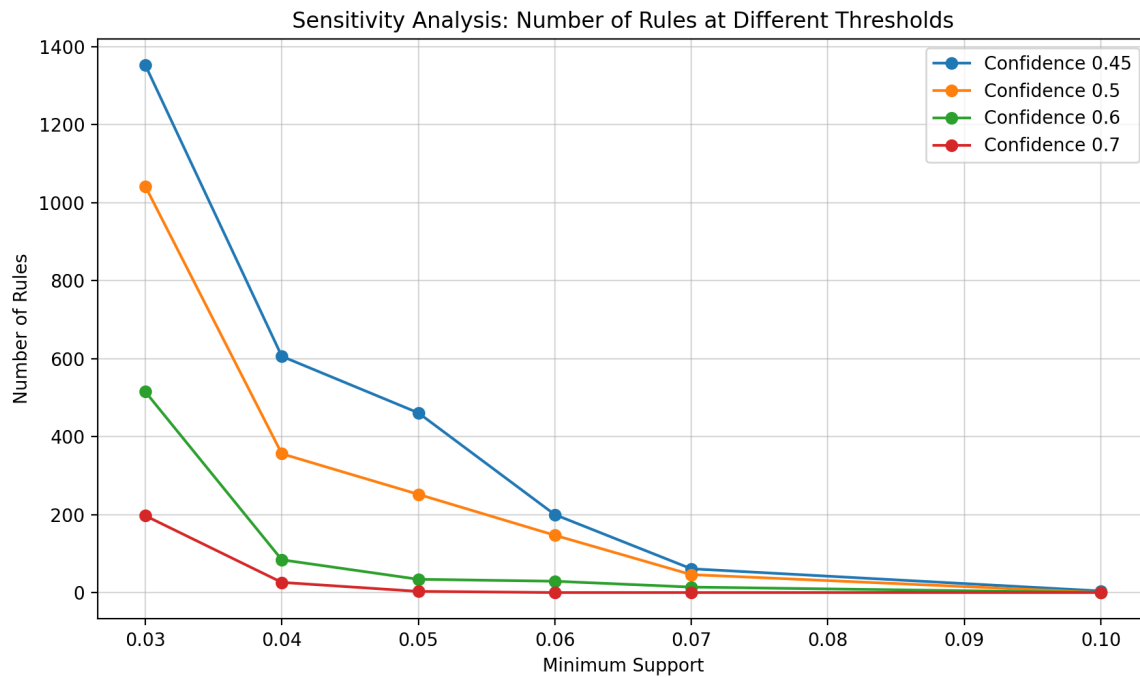


Figure: Sensitivity analysis of rule counts across support and confidence thresholds.

8. Country-Level Analysis

Country-level analysis was performed by identifying countries that appear most frequently in rows matching the strong rules. These countries can be prioritized for targeted environmental policy review.

Rank	Country	Strong Rule Appearances
1	Solomon Islands	14
2	Spain	10
3	Estonia	9
4	Hong Kong	9
5	Holy See (Vatican City State)	9
6	Afghanistan	9
7	Equatorial Guinea	8
8	Kenya	8
9	Angola	8
10	Tokelau	8

Targeted recommendations for the top 5 countries:

- Solomon Islands: Review recurring strong pollution-energy rule patterns, strengthen industrial waste monitoring, improve renewable energy planning, and assess whether energy recovery systems can reduce CO2 and pollution impacts.
- Spain: Review recurring strong pollution-energy rule patterns, strengthen industrial waste monitoring, improve renewable energy planning, and assess whether energy recovery systems can reduce CO2 and pollution impacts.
- Estonia: Review recurring strong pollution-energy rule patterns, strengthen industrial waste monitoring, improve renewable energy planning, and assess whether energy recovery systems can reduce CO2 and pollution impacts.

- Hong Kong: Review recurring strong pollution-energy rule patterns, strengthen industrial waste monitoring, improve renewable energy planning, and assess whether energy recovery systems can reduce CO2 and pollution impacts.
- Holy See (Vatican City State): Review recurring strong pollution-energy rule patterns, strengthen industrial waste monitoring, improve renewable energy planning, and assess whether energy recovery systems can reduce CO2 and pollution impacts.

9. Final Recommendations

- Invest in waste-to-energy systems, industrial heat recovery, and recycling-based energy recovery in countries with low or weak energy recovery patterns.
- Promote renewable energy sources such as solar, wind, hydro, and biomass in regions with high energy consumption.
- Strengthen industrial waste management through stricter monitoring, treatment, recycling, and cleaner production methods.
- Reduce fossil fuel dependency and improve energy efficiency in countries with high CO2 emission patterns.
- Use separate policies for air, water, and soil pollution because the rules show that pollution types do not always increase together.
- Use country-level rule appearances as a decision-support signal for targeted environmental interventions.

10. Limitations

The dataset contains only 200 rows. Although the strong rules were statistically significant within this dataset, a larger dataset would improve reliability and generalization. Association rules show relationships, not direct causation, so the results should be interpreted as decision-support patterns rather than causal proof.

11. Conclusion

This project successfully applied Apriori-based association rule mining to global pollution and energy data. After preprocessing and one-hot encoding, 34 strong statistically significant rules were discovered using support 0.05, confidence 0.60, lift greater than 1.0, and p-value less than 0.05. The results reveal meaningful patterns connecting pollution severity, industrial waste, CO2 emissions, energy recovery, energy consumption, and renewable energy usage. Sensitivity analysis confirmed that the selected thresholds were balanced, and country-level analysis provided targeted recommendations. Overall, association rule mining proved useful for discovering hidden pollution-energy relationships and supporting environmental decision-making.

Appendix: Compact Implementation Code

```
# Core implementation summary
from mlxtend.frequent_patterns import apriori, fpgrowth, association_rules
from scipy.stats import chi2_contingency

MIN_SUPPORT = 0.05
MIN_CONFIDENCE = 0.60

frequent_itemsets = apriori(encoded_df, min_support=MIN_SUPPORT, use_colnames=True)
rules = association_rules(frequent_itemsets, metric='confidence', min_threshold=MIN_CONFIDENCE)
rules = rules[rules['lift'] > 1.0]

chi2_results = []
for _, rule in rules.iterrows():
    ants = list(rule['antecedents'])
    cons = list(rule['consequents'])
    ant_true = encoded_df[ants].all(axis=1)
    con_true = encoded_df[cons].all(axis=1)
    n11 = (ant_true & con_true).sum()
    n10 = (ant_true & ~con_true).sum()
    n01 = (~ant_true & con_true).sum()
    n00 = (~ant_true & ~con_true).sum()
    chi2, p, dof, expected = chi2_contingency([[n11, n10], [n01, n00]])
    chi2_results.append({
        'antecedents': rule['antecedents'], 'consequents': rule['consequents'],
        'support': rule['support'], 'confidence': rule['confidence'],
        'lift': rule['lift'], 'p_value': p, 'significant': p < 0.05
    })

chi2_df = pd.DataFrame(chi2_results)
strong_rules = chi2_df[(chi2_df['confidence'] >= 0.60) &
                       (chi2_df['lift'] > 1.0) &
                       (chi2_df['p_value'] < 0.05)]
```